W-R-175/2D
22 December 1986
SUPERSEDING
W-R-175/2C
15 May 1973

FEDERAL SPECIFICATION SHEET

REEL, HUBS, FIBERGLASS AND METALLIC, WITH 3-INCH CENTER HOLE

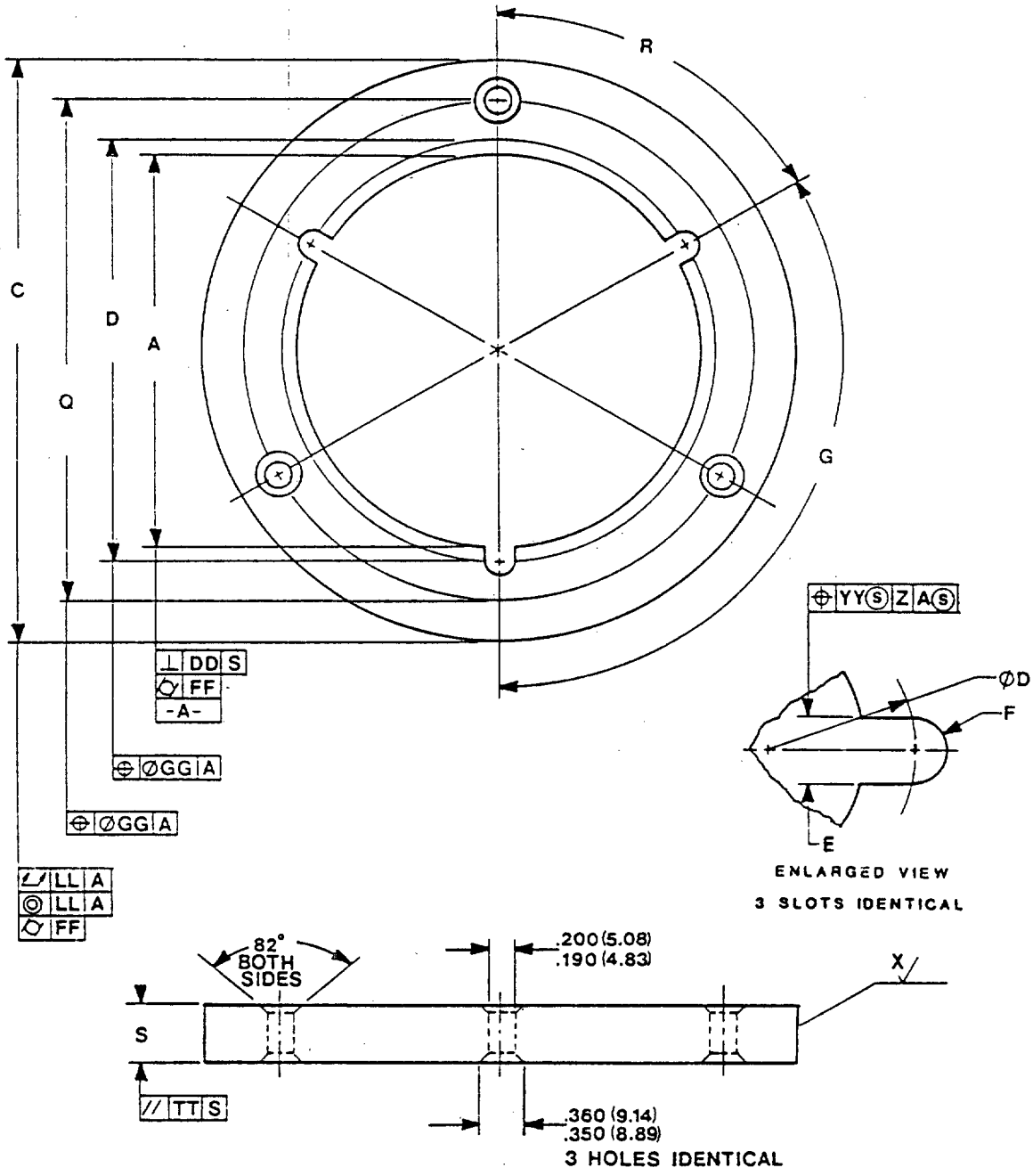
This specification sheet is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

The requirements for acquiring the hubs described herein shall consist of this specification and the latest issue of W-R-175.

AMSC N/A FSC 5835

DISTRIBUTION STATEMENT A Approved for public release; distribution unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.



SH 13202771

FIGURE 1. Hub configuration.

REQUIREMENTS:

1. Materials: "RB" hubs shall be constructed of fiberglass-filled plastic. "RC" hubs shall be constructed of anodized aluminum or an alloy with aluminum as the principal element.

2. Dimensions: The hubs shall be in accordance with the dimensions and measurements specified in table I (see figure 1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

TABLE I. Hub dimensions.[1] [2]

Characteristic[3]	Tape width indicator	Dimensions			
		"RB" hubs		"RC" hubs	
		maximum	minimum	maximum	minimum
Inside hub dia (A)		3.025 (76.835)	3.010 (76.454)	3.008 (76.403)	3.002 (76.251)
Outside hub dia (C)		4.510 (114.56)	4.490 (114.05)	4.510 (114.56)	4.490 (114.05)
Keyslot pitch dia (D)		3.270 (83.058)	3.248 (82.499)	3.258 (82.753)	3.248 (82.499)
Keyslot width (E)		0.229 (5.817)	0.219 (5.563)	0.229 (5.817)	0.219 (5.563)
Keyslot radius (F)		.5E	.5E	.5E	.5E
Angle between successive keyslots (degrees) (G)		120.25 (2.099rad)	119.75 (2.090rad)	120.25 (2.099rad)	119.75 (2.090rad)
Flange mounting[4] hole bolt circle diameter for for hubs (Q)		3.880 (98.552)	3.870 (98.298)	3.877 (98.476)	3.873 (98.374)
Angle between[4] successive keyslots and flange mounting holes for hubs (R)		60.5 (1.056rad)	59.5 (1.038rad)	60.26 (1.056rad)	59.75 (1.038rad)
Overall thickness (S) for hubs	250	0.355 (9.017)	0.345 (8.763)	0.355 (9.017)	0.345 (8.763)
	500	.605 (15.367)	.595 (15.113)	.605 (15.367)	.595 (15.113)
	750			.855 (21.717)	.845 (21.463)
	1000	1.105 (28.067)	1.095 (27.813)	1.105 (28.067)	1.095 (27.813)
	1500			1.605 (40.767)	1.595 (40.513)
	2000			2.105 (53.467)	2.095 (53.213)

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

TABLE I. Hub dimensions.[1] [2] - Continued

Characteristic[3]	Dimensions			
	"RB" hubs		"RC" hubs	
	maximum	minimum	maximum	minimum
Mounting surface parallelism (TT)	0.010 (0.254)		0.010 (0.254)	
Inside hub surface perpendicularity (DD)	.005 (0.127)		.005 (0.127)	
Inside hub surface cylindricity (FF)	.005 (0.127)		.002 (0.051)	
Outside hub surface cylindricity (FF)	.003 (0.076)		.002 (0.051)	
Outside hub surface concentricity (total runout) (LL)	.010 (0.254)		.010 (0.254)	
Keyslot position (YY)	.010 (0.254)		.010 (0.254)	
Roughness of outside hub surface (X)	125 microinches		80 microinches	
Number of flange mounting holes[4]				3
Diameter position (GG)	0.002 (0.051)		0.002 (0.051)	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

[1] Unless otherwise specified, all measurements are in inches. Metric equivalents are given for information only.
[2] All sharp edges shall be rounded to at least 0.0004 (0.102) radius.
[3] Where applicable, tolerance zones are indicated in parentheses.
[4] The use of flange mounting holes is optional for "RB" hubs.

QUALIFICATION:

1. The qualification sample for "RB" hubs shall consist of EH-250-RB hubs. The qualification sample for "RC" hubs shall consist of EH-250-RC hubs.

2. Qualification approval of EH-250-RC hubs shall constitute qualification for all "RC" hubs regardless of width.

3. Qualification approval of EH-250-RB hubs shall constitute qualification for all "RB" hubs regardless of width.

MILITARY INTERESTS:

Custodians

Army - CR
Navy - SH
Air Force - 99

Review activities

Navy - NV, OM
Air Force - 80
DLA - ES

User activity

Navy - AS

CIVIL COORDINATING ACTIVITY:
NASA - NPPO

Preparing activity:
Navy - SH
(Project 5835-0079-2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.